Magnetic Fields, $\vec{B}$

- Current produces Magnetic fields

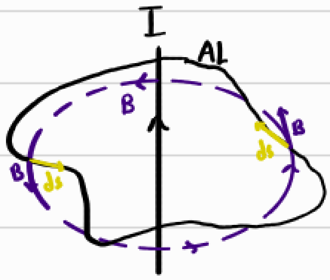
Ampere's Law

Amperian loop (AL)

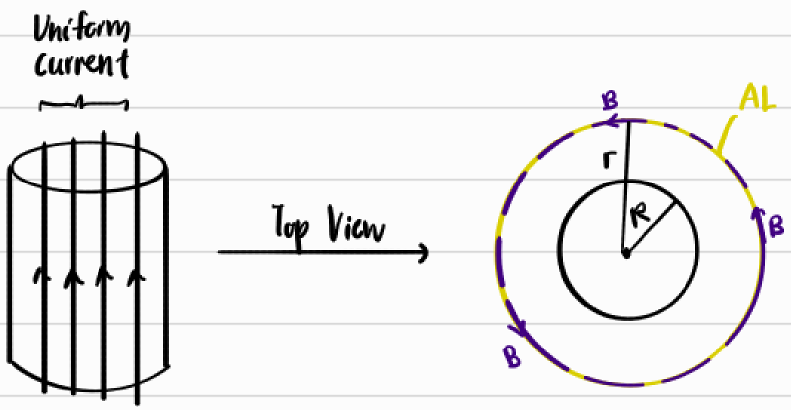
- Relationship btwn current enclosed by a closed loop w̄ the magnetic field along that loop

length vector — permiabality

$$\boxed{\oint \vec{B} \cdot \vec{ds} = \mu_0 I_{enclosed}}$$



* A straight current carrying wire produces concentric circular magnetic fields where the direc. of flow is determined by the Right Thumb Rule



Since B field are concentric circles, any circular AL will exactly overlay one of the infinately many B fields ∴ $\vec{B}$ will be tangent at every point of a circular AL

$$\oint \vec{B} \cdot \vec{ds} = \mu_0 I_{enclosed}$$

$$B \oint ds = \mu_0 I_{enclosed}$$

$$B(2\pi r) = \mu_0 I_{enclosed}$$

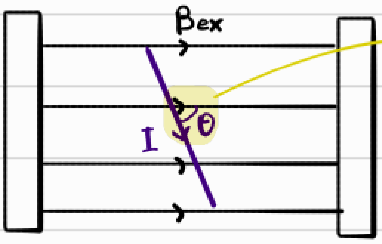
$$\boxed{B = \frac{\mu_0 I_{enclosed}}{2\pi r}}$$

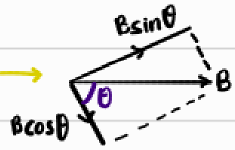
For solenoid,

$$\boxed{B = \mu_0 n I}$$

no. of loop per unit length

Force on current-carrying conductor in Magnetic Field





"$B_{ex}\cos\theta$" parallel to $\vec{I}$
"$B_{ex}\sin\theta$" perp. to $\vec{I}$

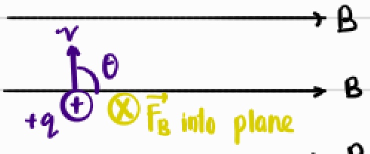
Let B_I = B by current-carrying conductor
B_{ex} = B by external sources

∴ There will only be a net force when B_{ex} is parallel w̄ B_I $\Longrightarrow$ Since $\vec{B_I} \perp \vec{I}$, to have a net force, $\vec{B_{ex}} \perp \vec{I}$, so that $\vec{B_{ex}}$ parallel w̄ $\vec{B_I}$
∴ $B_{ex}\sin\theta$ is the "useful component"

$$\boxed{\begin{aligned} \therefore\ |\vec{F_{BI}}| &= L\,(I B_{ex}\sin\theta) \\ \vec{F_{BI}} &= L\,(\vec{I} \times \vec{B_{ex}}) \end{aligned}}$$

Magnetic force on current-carrying conductor

Moving charges



$$F_B = BIL\sin\theta$$

$$F_B = B\left[\frac{Nq}{t}\right]L\sin\theta$$

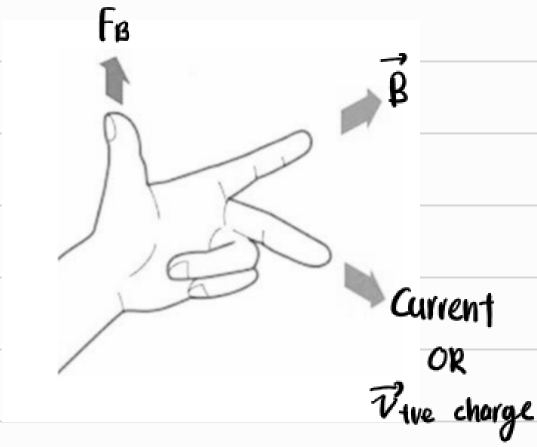
When $N = 1$,

$$F_B = Bq\left(\frac{L}{t}\right)\sin\theta$$

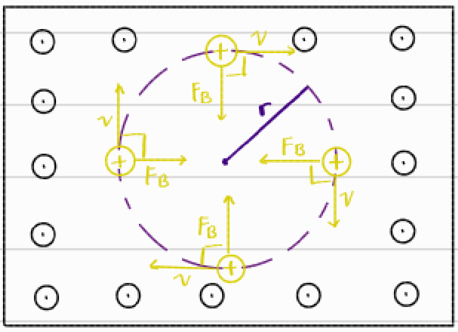
$$\boxed{F_B = Bqv\sin\theta}$$

$$\boxed{\vec{F}_B = (\vec{v} \times \vec{B})q}$$

Left Hand Rule



Uniform B-field



① $\vec{F}_B$ always perp. to $\vec{v}$ $[F_B = F_c]$

② $\vec{v}$ is constant

③ Since $F_B = F_c = \frac{mv}{r^2}$, F_c & v is constant

↳ r is constant

↳ Particle moves in circular motion

$$F_B = F_c$$

$$Bqv\sin 90^\circ = \frac{mv^2}{r}$$

$$\boxed{r = \frac{mv}{qB}}$$

constant

$$v = \frac{s}{t}$$

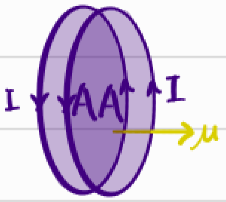
$$\boxed{T = \frac{2\pi r}{v}}$$

$$= \frac{2\pi}{v}\left(\frac{mv}{qB}\right)$$

$$\boxed{T = \frac{2\pi m}{qB}}$$

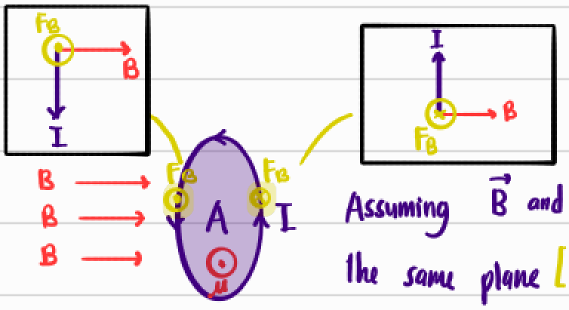
Magnetic Dipole Moment, μ

Circular conductor



$$\boxed{\mu = NIA}$$

[use Right Thumb Rule (solenoid version) to determine direc. of μ]



Assuming $\vec{B}$ and the loop of current is in the same plane [$\vec{B}$ is going hitting the loop, not through]

∴ These F_B causes torque on the loop to rotate horizontally until $\vec{\mu}$ is parallel to $\vec{B}$.

[As in the loop rotates with the right side of the loop going into the page]

Torque, $\tau = \vec{\mu} \times \vec{B}$

$$\boxed{\tau = \mu B \sin\theta}$$

As we can see, the higher the μ, the more torque the loop will experience.

[μ is a measure of how much a conductor reacts to an external magnetic field]